

From the knowledge domain perspective, recognition based AR requires deep understanding of computer vision and multimedia systems, especially that of video data (augmented reality works with continuous video feed from a camera; simple image recognition systems are not of much use here). Once again, all these disciplines depend on multiple mathematical research fields. If the library you are working with does not provide you with the required features, you are going to need superb mathematical skills to get things to work your way.

Outlining based AR

Outlining based AR heavily depends on object recognition and hence, a significant portion of knowledge required to work with recognition based systems is also applicable to the outlining based AR systems. However the game is not all the same. Outlining based AR applications may use data sources from other sensors too, such as compass, GPS, accelerometers, ultrasound sensors, infrared sensors and so on. Depending on the requirements of the particular application you want to develop, conceptual and domain knowledge of one or more of the above will be necessary.

Software required

Let us say you want to build a location based AR browser. To achieve the feat, you need to plan first and for that, you would need answers to a few questions. We try to show you some which you would most certainly have to face:

► What platform should I build it for?

If you are using an Android phone, you would target android; if you are an iPhone user, it'd be iOS. If you want to build a commercial application, you would want to target them both and may be others too, such as Blackberry and Windows Phones. The target platform depends on who your users are and need you wish to fulfil.

► What programming languages can I use?

That would depend on the platform that you are targeting. If you want Android, you need Java. Objective C if it's for the iPhone while Windows and Blackberry phones allow you to get creative with multiple languages.

► What OS can I use as a development platform?

Once again, it depends on your target platform. Windows phone apps